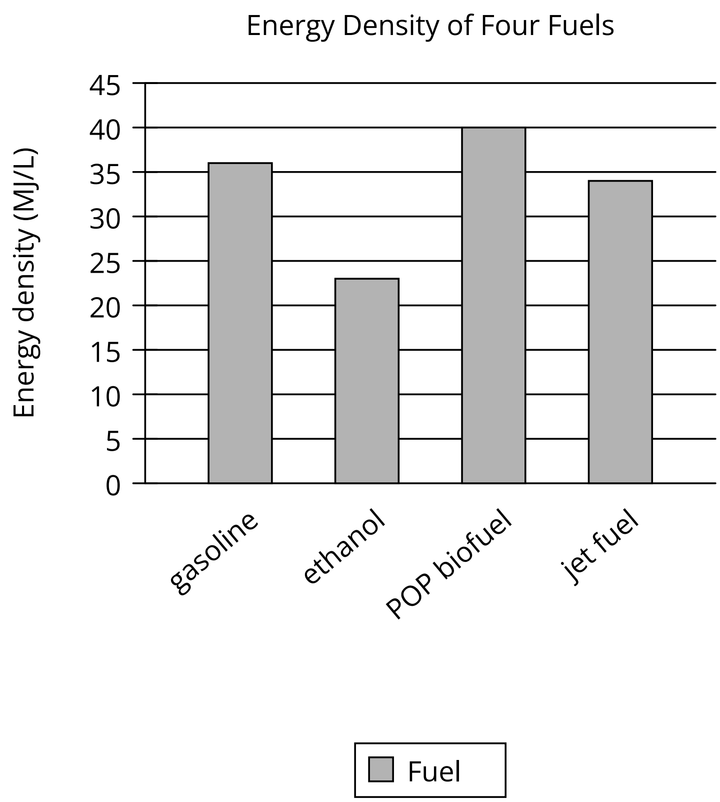
Question ID: b29c520a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Easy

Question



A team of researchers used bacteria to create a biofuel (a renewable fuel produced from plants, algae, or other living materials). The team called the new fuel POP biofuel. To determine how much energy is stored in POP biofuel compared to other fuels, the team calculated the fuels' energy density, in megajoules per liter (MJ/L). The team found that the fuel with the highest energy density is ______

Which choice most effectively uses data from the graph to complete the sentence?

Answer

A. jet fuel.

B. POP biofuel.

C. ethanol.

D. gasoline.

Correct Answer: B

Rationale

Choice B is the best answer because it most effectively uses data from the graph to complete the sentence about which fuel has the highest energy density. Of the four bars in the graph, the highest (indicating the greatest energy density in MJ/L) is for POP biofuel. According to the graph, the energy densities of the fuels shown are approximately 40 MJ/L for POP biofuel, 36 MJ/L for gasoline, 34 MJ/L for jet fuel, and 23 MJ/L for ethanol.

Choice A is incorrect. The task is to find the fuel in the graph with the highest bar (indicating the greatest energy density in MJ/L), and jet fuel has the third-highest bar in the graph, not the highest. According to the graph, the energy densities of the fuels shown are approximately 40 MJ/L for POP biofuel, 36 MJ/L for gasoline, 34 MJ/L for jet fuel, and 23 MJ/L for ethanol. Choice C is incorrect. The task is to find the fuel in the graph with the highest bar (indicating the greatest energy density in MJ/L), and ethanol has the lowest bar in the graph, not the highest. According to the graph, the energy densities of the fuels shown are approximately 40 MJ/L for POP biofuel, 36 MJ/L for gasoline, 34 MJ/L for jet fuel, and 23 MJ/L for ethanol. Choice D is incorrect. The task is to find the fuel in the graph with the highest bar (indicating the greatest energy density in MJ/L), and gasoline has the second-highest bar in the graph, not the highest. According to the graph, the energy densities of the fuels shown are approximately 40 MJ/L for POP biofuel, 36 MJ/L for gasoline, 34 MJ/L for jet fuel, and 23 MJ/L for ethanol.